Introduction to Biotechnology

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Biotechnology is the use of microbes, animal/plant cells and their products to synthesize, break down or transform materials. Primarily it includes the use of recombinant DNA technology and genetic

Introduction to Biotechnology

engineering techniques to improve upon the quality of processes.

Traditional biotechnology refers to the conventional techniques that have been used for many

Introduction to Biotechnology

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centuries to produce beer, wine, cheese etc.



Introduction to Biotechnology

Modern Biotechnology embraces all methods of genetic modification by recombinant DNA & cell fusion techniques together with



Introduction to Biotechnology

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the modern developments of traditional biotechnological processes.

Branches of Biotechnology

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White Biotechnology

Development of processes and microorganisms for Industrial processes.

Key concept

Scale of operation

Low volume high price

Branches of Biotechnology

White Biotechnology

Example

- Enzyme production



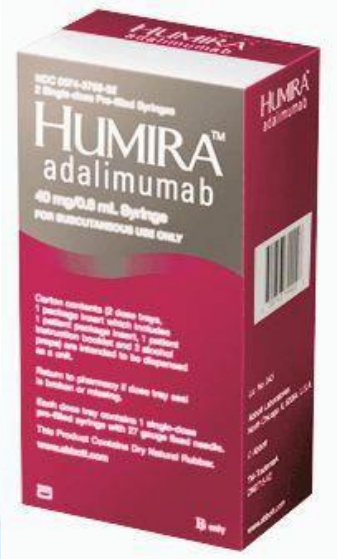
Branches of Biotechnology

https://chat.whatsapp.com/EEkLcDyZoYr5wdtScC3W3t?mode=wwt

Red Biotechnology

It is concerned with the discovery and development of innovative drugs and treatments.

Branches of Biotechnology



Branches of Biotechnology

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Green Biotechnology

It is concerned with the modification of the genetic composition of plants to enhance existing traits or add new ones.

Branches of Biotechnology

Green Biotechnology

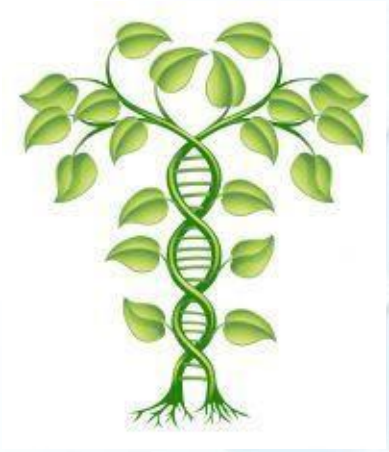
Example

Bt corn from Syngenta

Bt cotton from Monsanto

Branches of Biotechnology

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Modern Biotechnology
(Principles & Applications)

Course Credits 3 **Course**

Instructor

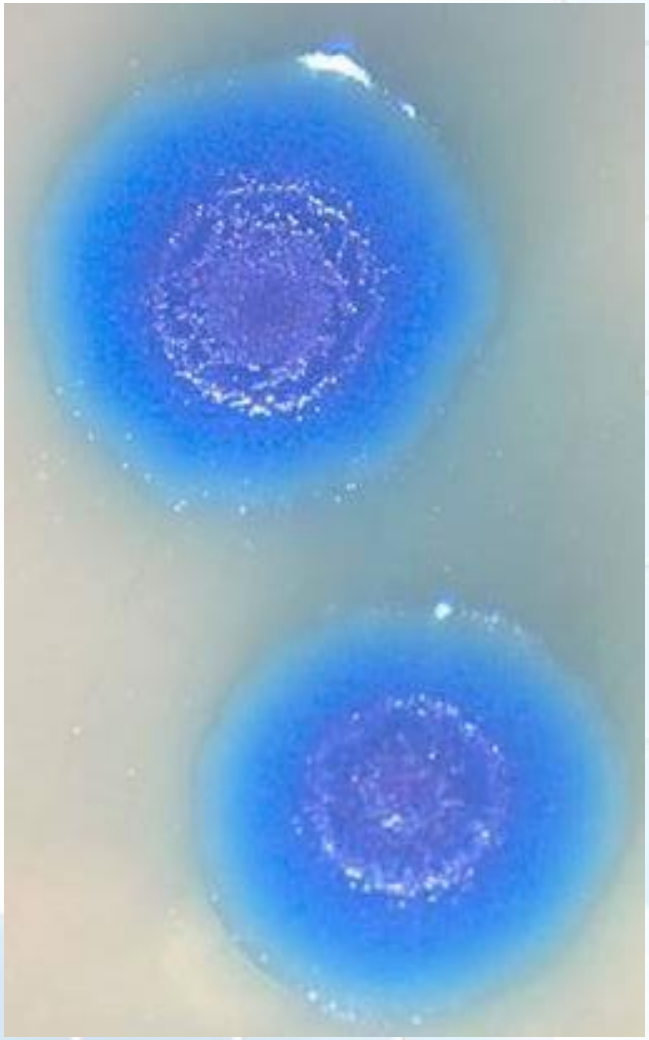
Dr. Muhammad Safwan Akram

Ph.D. Biotechnology
(Univ. of Cambridge, UK)
M.Phil Bioscience Enterprise
(Univ. of Cambridge, UK)

Biotechnology in the 21st Century

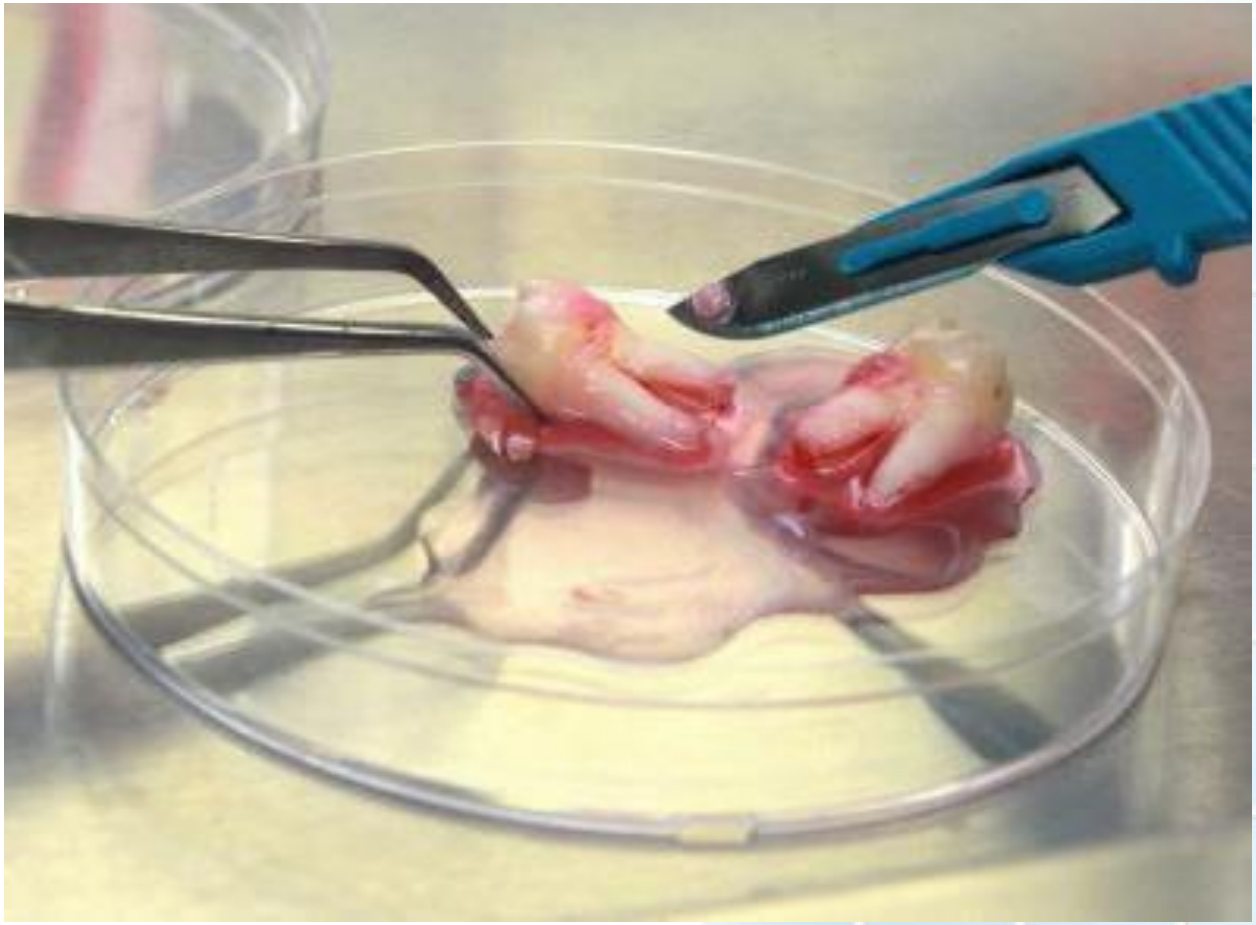
M.Sc. Biotechnology
(*Univ. of the Punjab,*
Lahore,
Pakistan)
B.Sc. (Hons.) Biochemistry
(Univ. of the Punjab,
Lahore,
Pakistan)

Biotechnology in the 21st Century

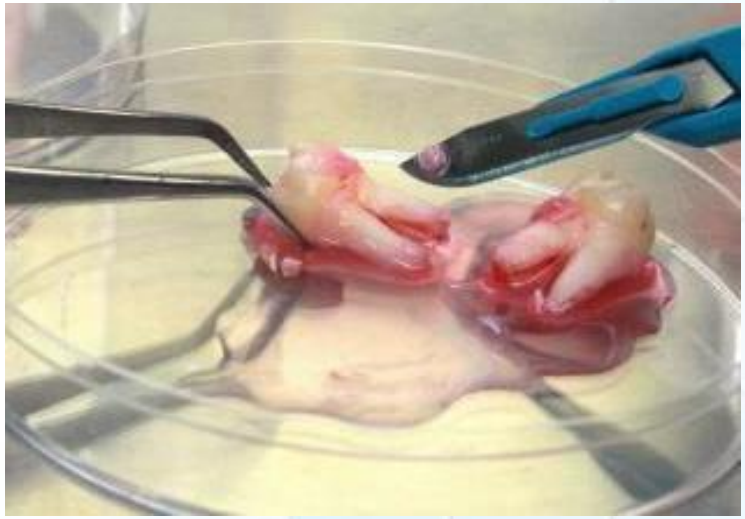


Biotechnology in the 21st Century

Growing Teeth



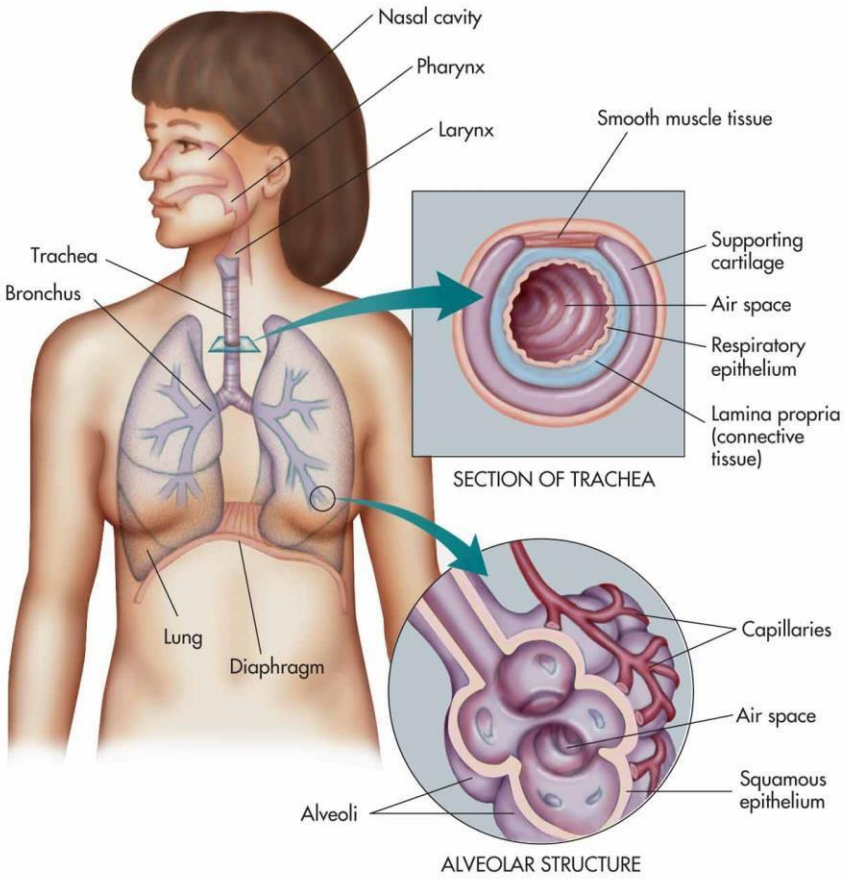
Biotechnology in the 21st Century



Biotechnology in the 21st Century

Claudia's Trachea

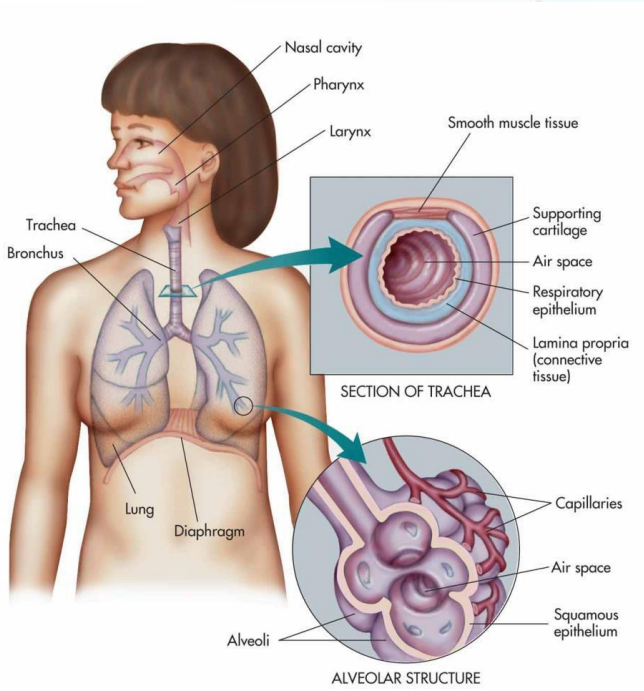
Biotechnology in the 21st Century







Biotechnology in the 21st Century

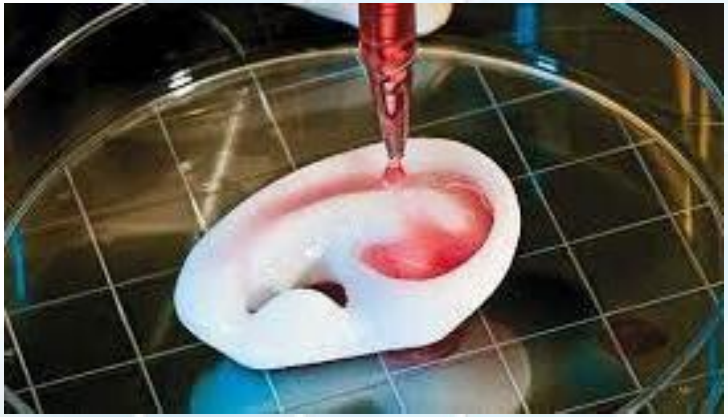




Biotechnology in the 21st Century

Growing Organs





Biotechnology in the 21st Century

Growing Blood Vessels



Chicken with an extra leg

Biotechnology in the 21st Century

HARVARD MEDICAL SCHOOL





Growing Meat

Biotechnology in the 21st Century



Roots of Biotechnology

6000 BC

Sumerians & Babylonians used **anaerobic respiration** of yeast to make beer.



Roots of Biotechnology

4000 BC

Egyptians used **aerobic respiration** of yeast to leaven or rise bread and



used mold to flavour cheese.



Roots of Biotechnology

1350 BC

Pregnancy test

Barley grows – male child Wheat grows – female child



1350 BC

Pregnancy test

Barley grows – male child

Roots of Biotechnology

Wheat grows – female child



400 BC

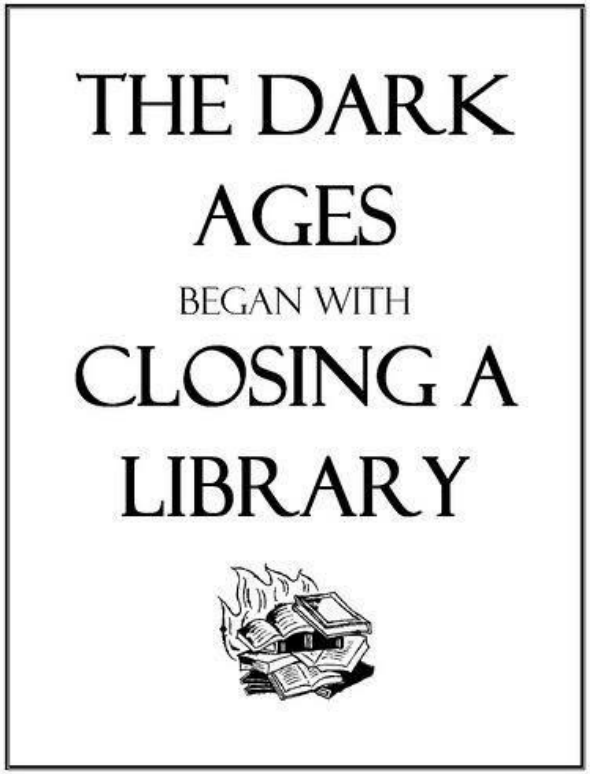
Use of salts to preserve food specially meat.

Roots of Biotechnology



European Dark Period &
Islamic Golden Era

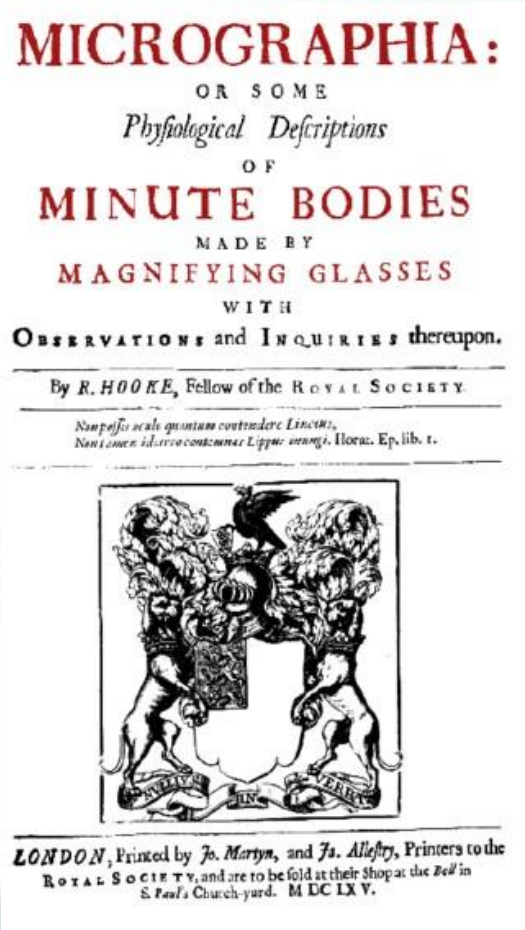
Roots of Biotechnology



Roots of Biotechnology

1665 AD

Roots of Biotechnology



MICROGRAPHIA:

OR SOME

Physiological Descriptions

OF

MINUTE BODIES

MADE BY

MAGNIFYING GLASSES

WITH

OBSERVATIONS and INQUIRIES thereupon.

By *R. HOOKE*, Fellow of the ROYAL SOCIETY.

Nam possis oculis quantum contendere Lynceus,
Non tamen idcirco contemnas Lippus inungi. Horat. Ep. lib. 1.

LONDON, Printed by *Jo. Martyn*, and *Ja. Allestry*, Printers to the ROYAL SOCIETY, and are to be sold at their Shop at the *Bell* in S. *Paul's* Church-yard. M DC LX V.

Roots of Biotechnology

1676 AD

Antonie Van Leeuwenhoek

Observation of microscopic single celled organisms

Roots of Biotechnology



1828 AD

Synthesis of Urea

Roots of Biotechnology



Fredrick Wohler

Pioneer in inorganic chemistry for synthesizing urea. This discovery became refutation of vitalism, the hypothesis that living things are alive because of some special "vital force"

Roots of Biotechnology

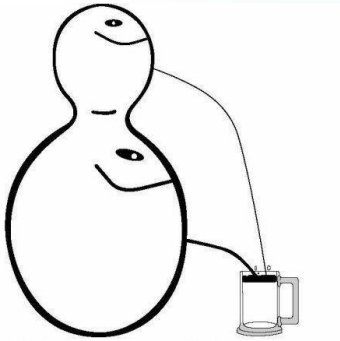
1834 - 1839 AD

Cell Theory

Theodore Schwann

"All living things are composed of cells and cell products"

Roots of Biotechnology



1857 AD

Rudolf Virchow

Omnis cellula e cellula

"Every cell arise from another cell"

Roots of Biotechnology



End

Roots of Biotechnology (2)

1849

Industrial production of Baker's yeast



Roots of Biotechnology (2)

1876

Kuhne creates the term "enzymes" and discovered trypsin.

Roots of Biotechnology (2)

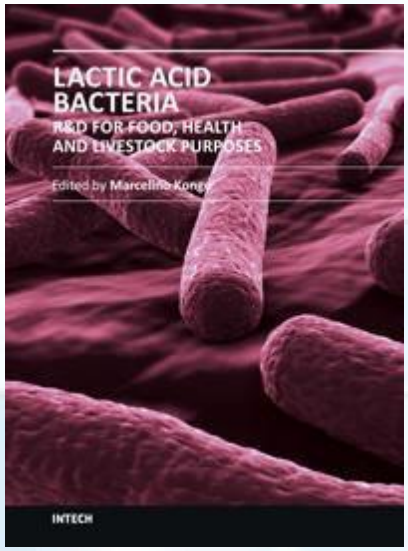


Roots of Biotechnology (2)

1881

Industrial microbial production of lactic acid from lactic acid bacteria

Roots of Biotechnology (2)



Roots of Biotechnology (2)

1890

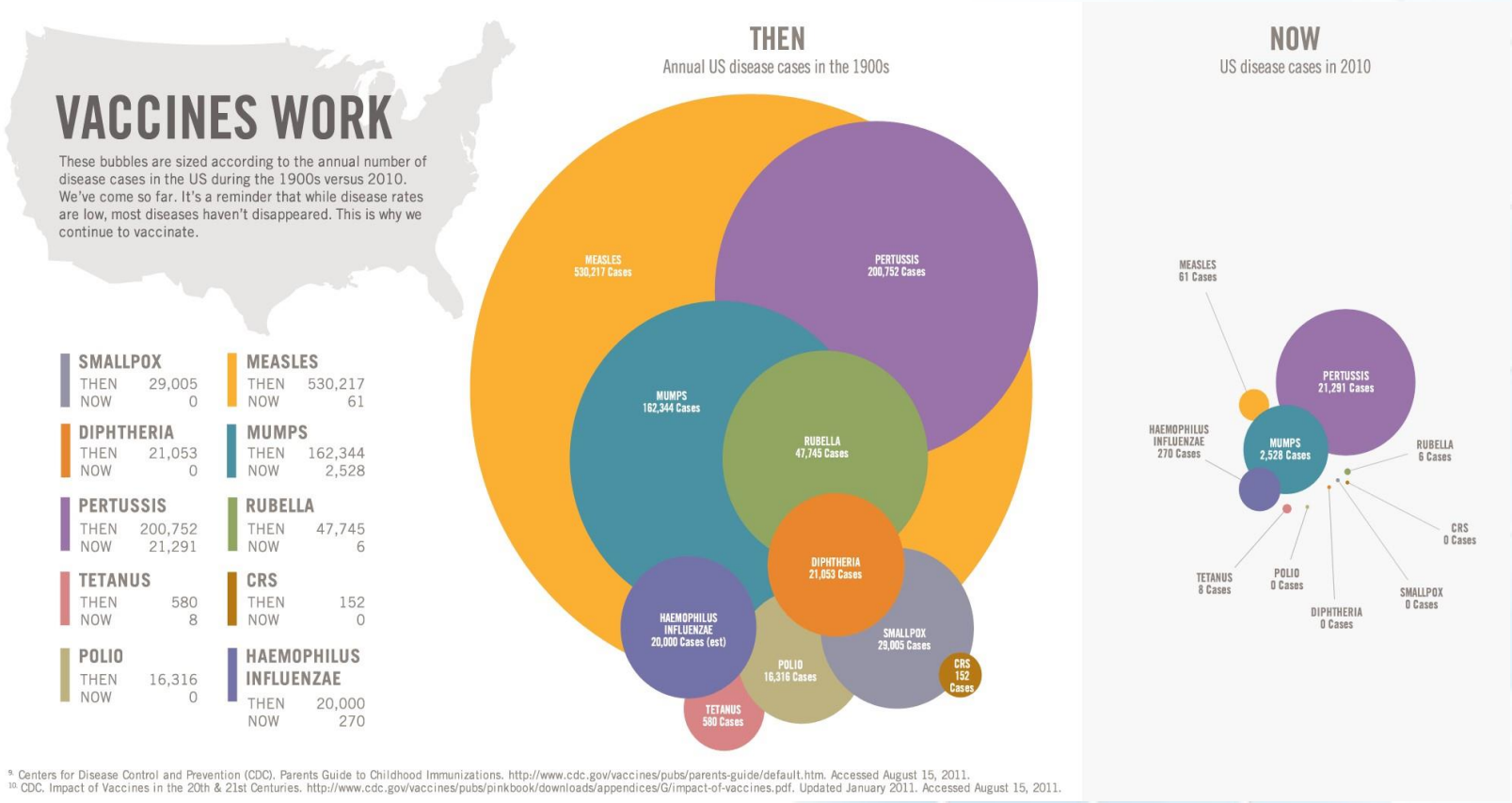
Development of first vaccines by Robert Koch and Louis Pasteur.

Roots of Biotechnology (2)





Roots of Biotechnology (2)



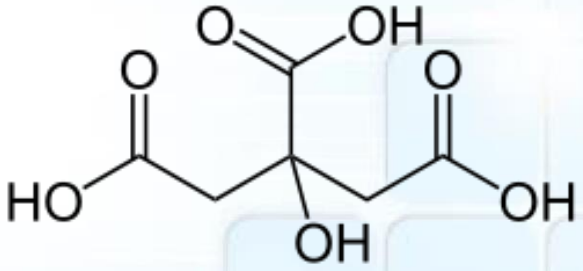
[9] Centers for Disease Control and Prevention (CDC). Parents Guide to Childhood Immunizations. http://www.cdc.gov/vaccines/pubs/parents-guide/default.htm. Accessed August 15, 2011.
[10] CDC. Impact of Vaccines in the 20th & 21st Centuries. http://www.cdc.gov/vaccines/pubs/pinkbook/downloads/appendices/G/impact-of-vaccines.pdf. Updated January 2011. Accessed August 15, 2011.

Roots of Biotechnology (2)

1893

Industrial production of citric acid by Aspergillus Niger

Roots of Biotechnology (2)





Roots of Biotechnology (2)

1897

Eduard Buchner made cell free extract of yeast cells and showed that it could still ferment sugar.

Roots of Biotechnology (2)



1915

Patent granted for enzymes in the detergent industry and leather

Roots of Biotechnology (2)



Production of Cordite

production to Otto Rohm
1917

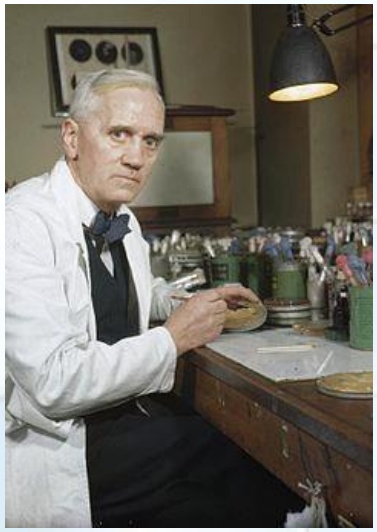
Industrial fermentation process for Acetone, Butanol and Ethanol **from starch.**

Roots of Biotechnology (2)

Chaim Weizmann
1928

Alexander Fleming discovers penicillin Penicillium Notatum

Roots of Biotechnology (2)



Discovery of DNA

1869

Friedrich Miescher

discovered nuclein

1878

Albrecht Kossel

Isolated non-protein component of nuclein & discovered nucleic acids & isolated 5 bases: A, T,

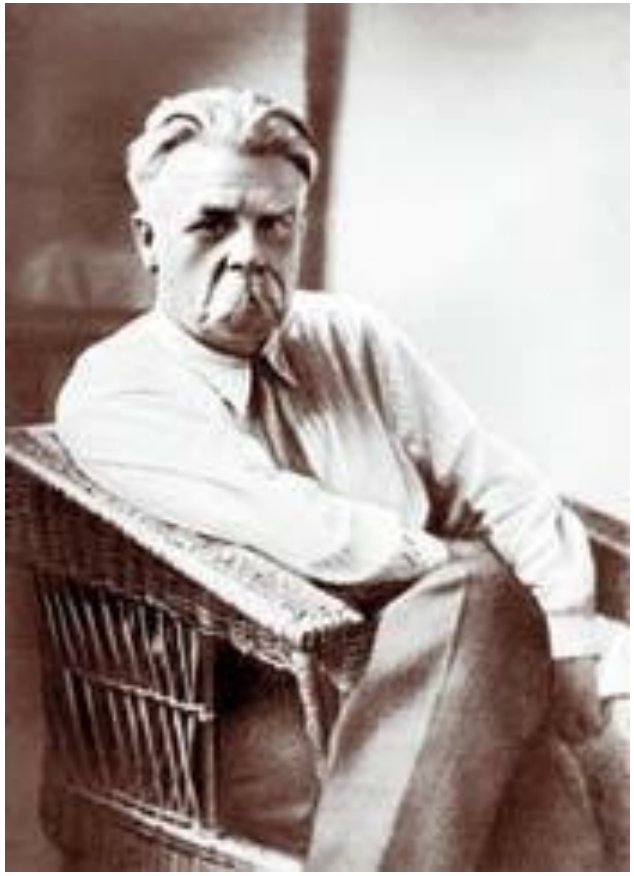
Discovery of DNA

G, C & U

Discovery of DNA

1927

Discovery of DNA



Nikolai Konstantinovich Koltsov

Inherited traits would be inherited via a "giant hereditary molecule" which would be made up of "two mirror strands that would replicate in a semiconservative fashion

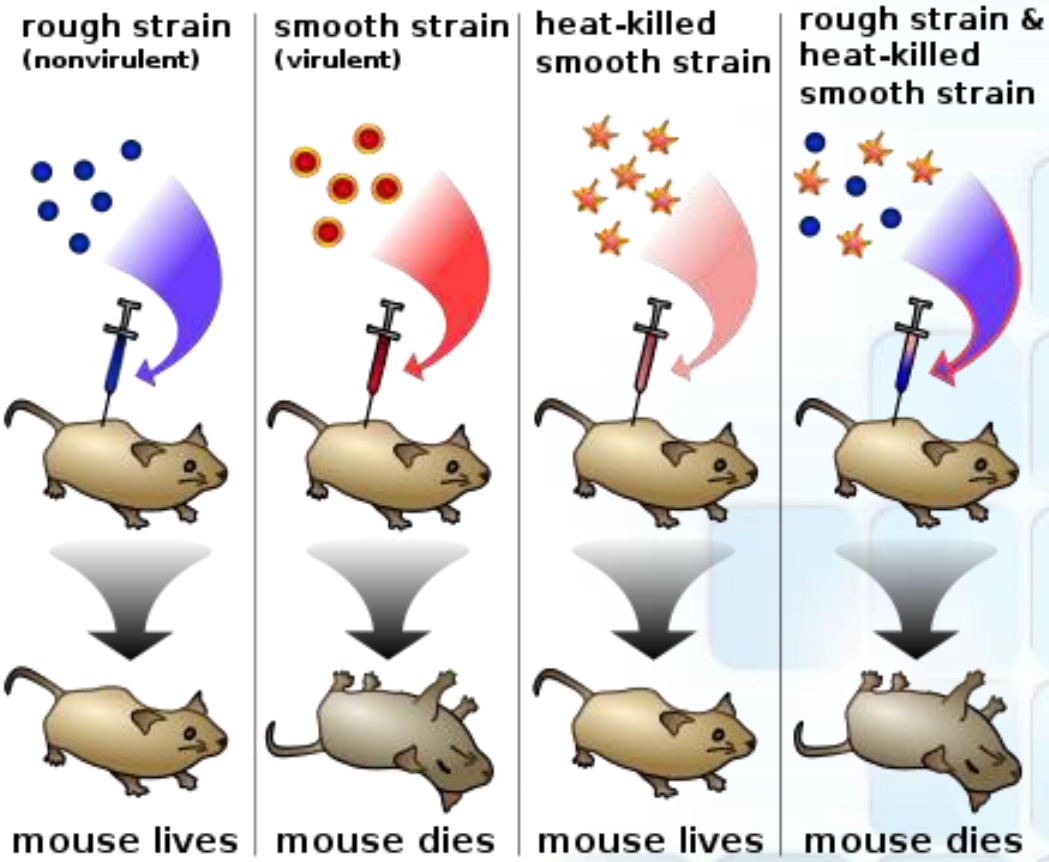
Discovery of DNA

using each strand as a template"

1928

Frederick Griffith

Discovery of DNA



Discovery of DNA

1944

Avery, MacLeod & McCarty Experiment

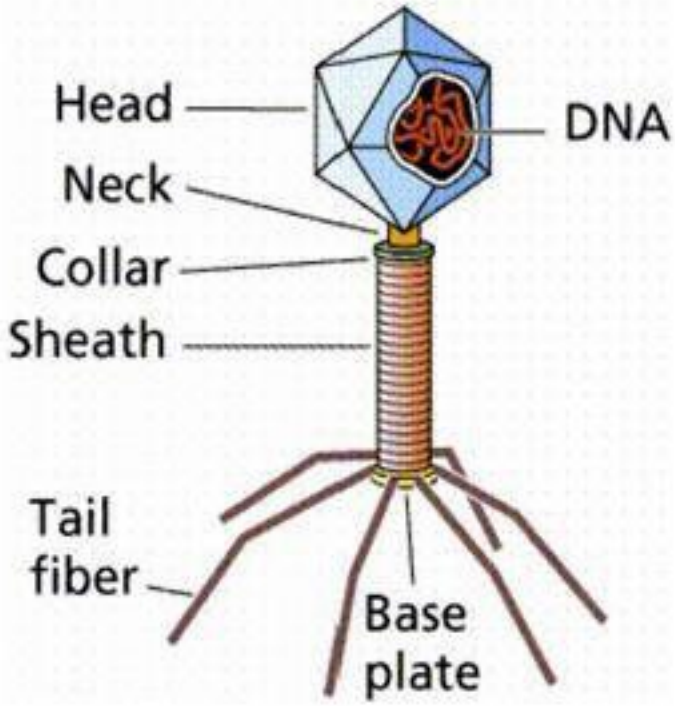
Discovery of DNA



1952

Hershey-Chase Experiment

Discovery of DNA



1952

Hershey-Chase Experiment

Discovery of DNA

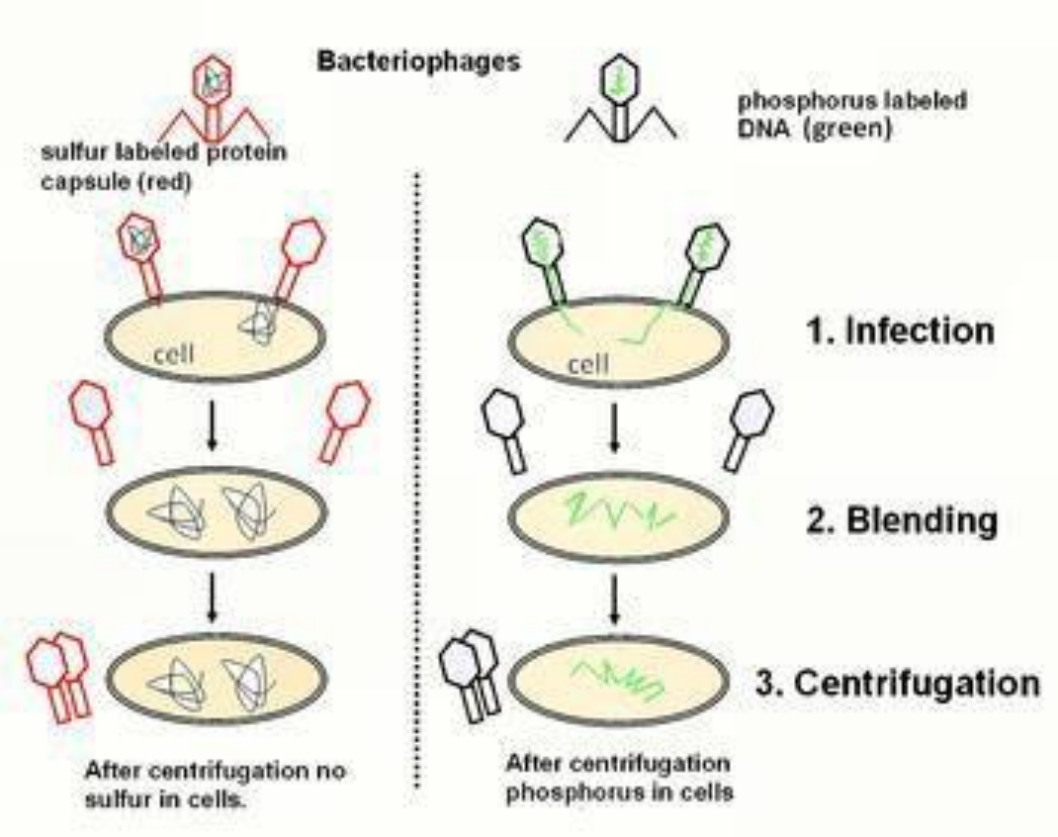
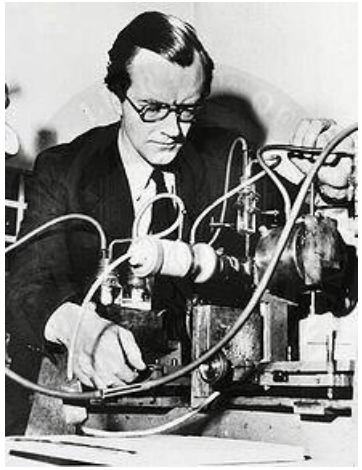


Photo 51

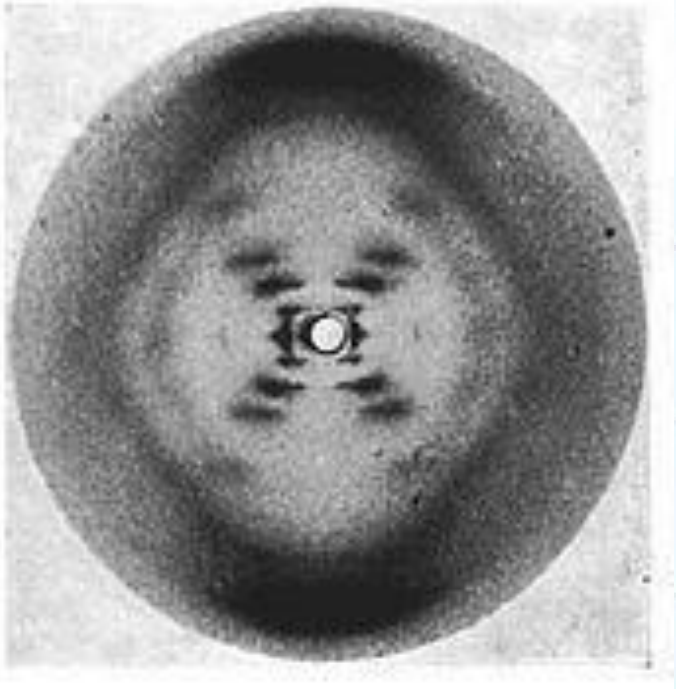
Discovery of DNA

Rosalind Franklin



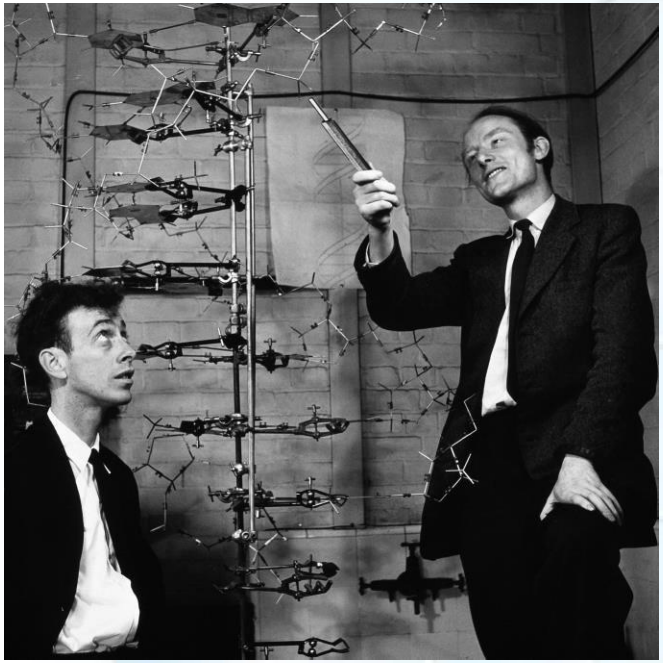


Maurice Wilkins



Watson & Crick 1953

Discovery of DNA



Discovery of DNA

MOLECULAR STRUCTURE OF NUCLEIC ACIDS

A Structure for Deoxyribose Nucleic Acid

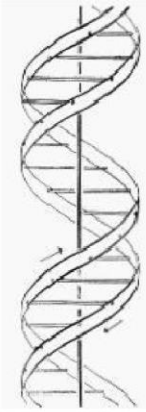
We wish to suggest a structure for the salt of deoxyribose nucleic acid (D.N.A.). This structure has novel features which are of considerable biological interest.

A structure for nucleic acid has already been proposed by Pauling and Corey[1]. They kindly made their manuscript available to us in advance of publication. Their model consists of three intertwined chains, with the phosphates near the fibre axis, and the bases on the outside. In our opinion. this structure is unsatisfactory for two reasons:

(1) We believe that the material which gives the X-ray diagrams is the salt, not the free acid. Without the acidic hydrogen atoms it is not clear what forces would hold the structure together, especially as the negatively charged phosphates near the axis will repel each other. (2) Some of the van der Waals distances appear to be too small.

Another three-chain structure has also been suggested by Fraser (in the press). In his model the phosphates are on the outside and the bases on the inside, linked together by hydrogen bonds. This structure as described is rather ill-defined, and for this reason we shall not comment on it.

We wish to put forward a radically different structure for the salt of deoxyribose nucleic acid. This structure has two helical chains each coiled round the same axis (see diagram). We have made the usual chemical assumptions. namely, that each chain consists of phosphate diester groups joining β-D-deoxyribofuranose residues with 3',5' linkages. The two chains (but not their bases) are related by a dyad perpendicular to the fibre axis. Both chains follow righthanded helices, but owing to the dyad the sequences of the atoms in the two chains run in opposite directions.



This figure is purely diagrammatic. The two ribbons symbolize the two phosphate—sugar chains, and the horizontal rods the pairs of bases holding the chains together. The vertical line marks the fibre axis

Each chain loosely resembles Furberg's[2] model No. 1; that is, the bases are on the inside of the helix and the phosphates on the outside. The configuration of the sugar and the atoms near it is close to Furberg's 'standard configuration', the sugar being roughly perpendicular to the attached base. There is a residue on each chain every 3·4 A. in the z-direction. We have assumed an angle of 36° between adjacent residues in the same chain, so that the structure repeats after 10 residues on each chain, that is, after 34 A. The distance of a phosphorus atom from the fibre axis is 10 A. As the phosphates are on the outside, cations have easy access to them.

The structure is an open one, and its water content is rather high. At lower water contents we would expect the bases to tilt so that the structure could become more compact.

The novel feature of the structure is the manner in which the two chains are held together by the purine and pyrimidine bases. The planes of the bases are perpendicular to the fibre axis. They are joined together in pairs, a single base from one chain being hydrogen-bonded to a single base from the other chain, so that the two lie side by side with identical z-co-ordinates. One of the pair must be a purine and the other a pyrimidine for bonding to occur. The hydrogen bonds are made as follows: purine position 1 to pyrimidine position 1; purine position 6 to pyrimidine position 6.

If it is assumed that the bases only occur in the structure in the most plausible tautomeric forms (that is, with the keto rather than the enol configurations) it is found that only specific pairs of bases can bond together. These pairs are: adenine (purine) with thymine (pyrimidine), and guanine (purine) with cytosine (pyrimidine).

In other words, if an adenine forms one member of a pair, on either chain, then on these assumptions the other member must be thymine; similarly for guanine and cytosine. The sequence of bases on a single chain, does not appear to be restricted in any way. However, if only specific pairs of bases can be formed, it follows that if the sequence of bases on one chain, is given, then the sequence on the other chain is automatically determined.

It has been found experimentally[3,4] that the ratio of the amounts of adenine to thymine, and the ratio of guanine to cytosine. are always very close to unity for deoxyribose nucleic acid.

It is probably impossible to build this structure with a ribose sugar in place of the deoxyribose, as the extra oxygen atom would make too close a van der Waals contact.

The previously published X-ray data[5,6] on deoxyribose nucleic acid are insufficient for a rigorous test of our structure. So far as we can tell, it is roughly compatible with the experimental data, but it must be regarded as unproved until it has been checked against more exact results. Some of these are given in time following, communications. We were not aware of the details of the results presented there when we devised our structure, which rests mainly though not entirely on published experimental data and stereo-chemical arguments.

It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material.

Full details of the structure, including the conditions assumed in building it, together with a set of co-ordinates for the atoms, will be published elsewhere.

We are much indebted to Dr. Jerry Donohue for constant advice and criticism, especially on interatomic distances. We have also been stimulated by a knowledge of the general nature of the unpublished experimental results and ideas of Dr. *M.* H. F. Wilkins, Dr. R. E. Franklin and their co-workers at King's College, London. One of us (J.D.W.) has been aided by a fellowship from the National Foundation for Infantile Paralysis.

J.D. WATSON
F.H. C. CRICK

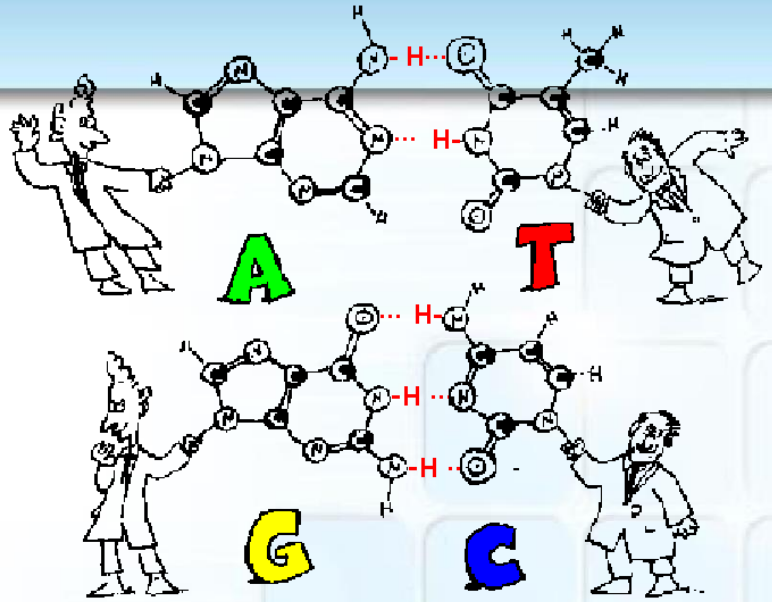
Medical Research Council Unit for the Study of the Molecular Structure of Biological Systems, Cavendish Laboratory, Cambridge. April 2.

[1] Pauling, L., and Corey, R. B. nature, 171, 346 (1953); Proc, U.S. Nat Acad. Sci., 39, 84 (1953).
[2] Furberg, S., Acta Chem. Scand., 6, 634 (1952).
[3] Chargaff, E., for references see Zamenhof, S., Brawerman, G., and Chargaff, E., Biochim, et Biophys. Acta, 9 402 (1952).
[4] Wyatt, G.R. J. Gen. Physiol., 36 201 (1952).
[5] Astbury, W.T., Symp. Soc. Exp. Biol. 1, Nucleic Acid, 66 (Camb. Univ. Press, 1947)
[6] Wilkins, M. H. F. and Randall, J. T. Biochim. et. Biophys. Acta, 10, 192 (1953).

Dawn of Biotechnology

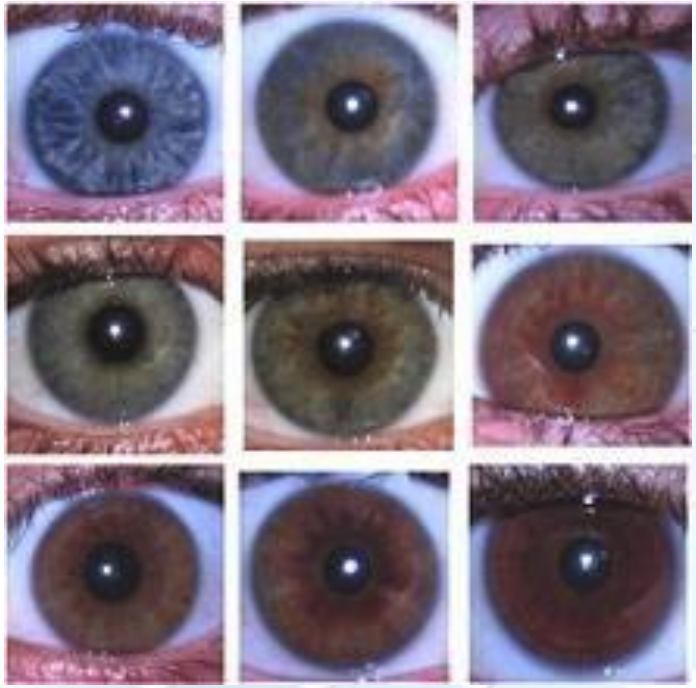
Discovery of DNA

Watson & Crick 1953


A
T
G
C

Dawn of Biotechnology

Spectrum of Eye Colour



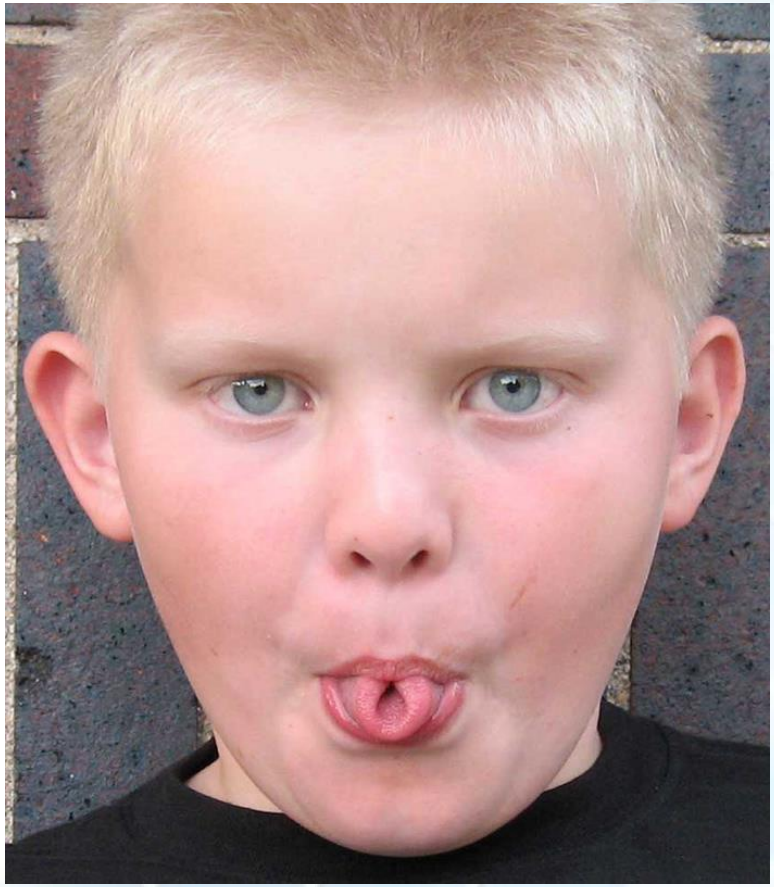
Dawn of Biotechnology

Curly hair



Tongue Rolling

Dawn of Biotechnology



Can we control the traits?

Dawn of Biotechnology

Can we isolate the genes?

Can we edit the genes?

Can we introduce them in new species?

Freckles



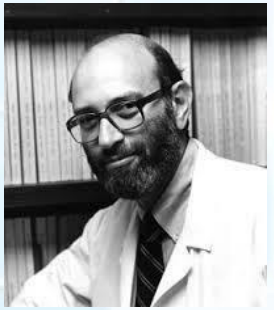
Dawn of Biotechnology

Restriction Endonucleases









Dawn of Biotechnology

1973

- Boyer brought in restriction enzymes
- Cohen contributed with plasmid (pSC101)
- African Clawed Frog gene into a bacterium

Dawn of Biotechnology

Negative Perception

"Genetic Engineering can give rise to something which can crawl out of the laboratory such as Frankenstein"

Dawn of Biotechnology



Dawn of Biotechnology







Dawn of Biotechnology

Genentech
A Member of the Roche Group

KPCB | KLEINER PERKINS CAUFIELD BYERS

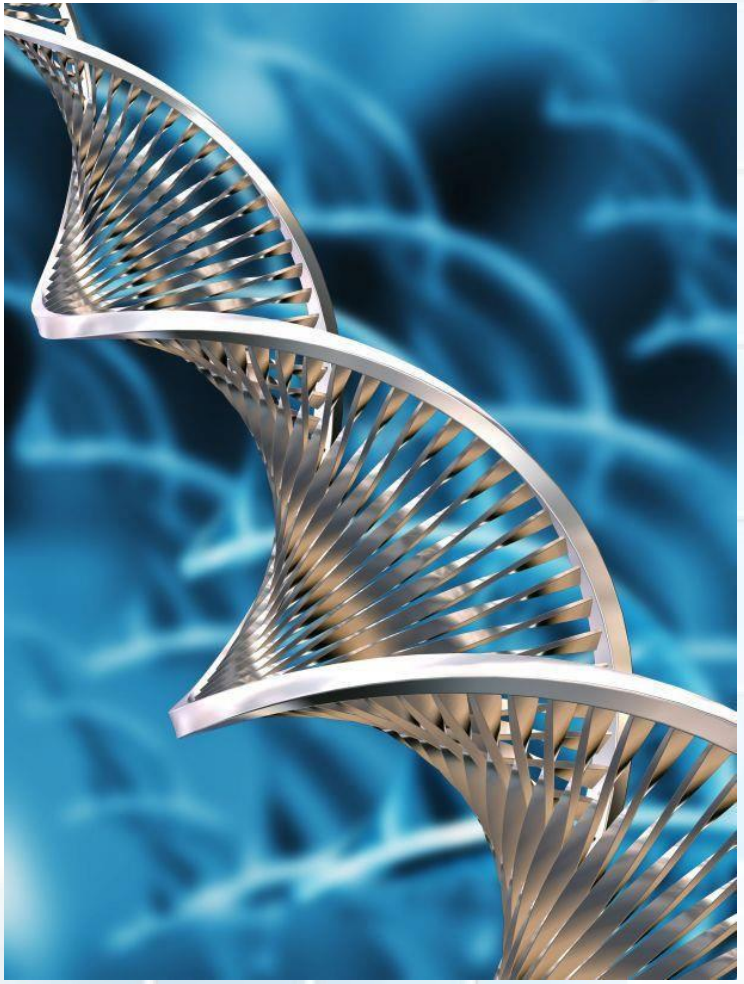


Dawn of Biotechnology





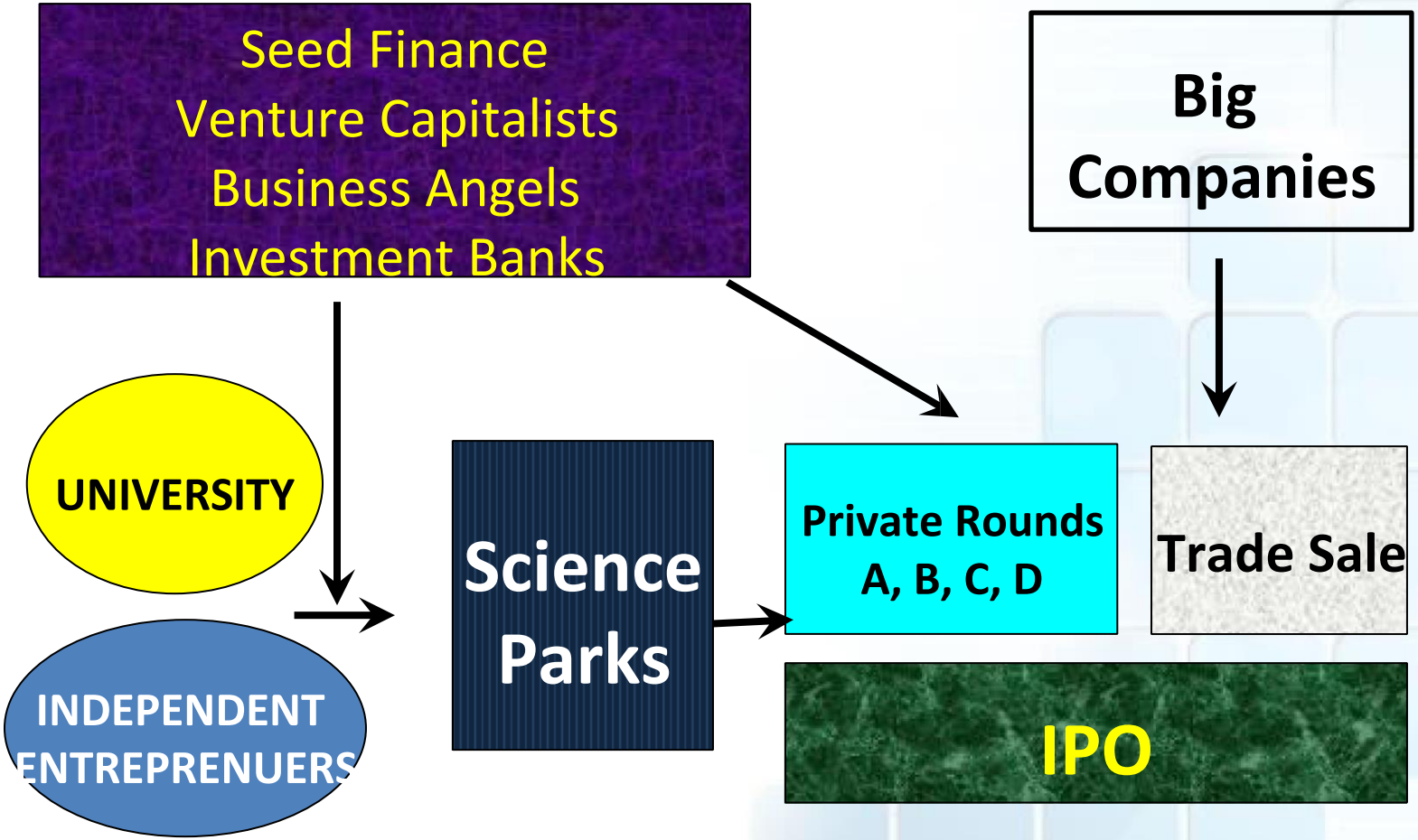
Commercializing Biotechnology



Commercializing Biotechnology

Company Formation In Biotechnology

Commercializing Biotechnology



Commercializing Biotechnology

First generation of Biotech Companies

Commercializing Biotechnology

Company	Date of IPO	Amount Raised
Genentech	10/80	$35M
Cetus	03/81	$107M
Genetic Systems	04/81	$6M
Ribi Immunochem	05/81	$1.8M
Genome Therapeutics	O5/82	$12.9M
Centocor	12/82	$21M
Scios	01/83	$12M

Commercializing Biotechnology

First generation of Biotech Companies

Commercializing Biotechnology

Company	Date of IPO	Amount Raised
Biotechnology General	09/83	$8.9M
Immunex	03/83	$16.5M
Amgen	06/83	$42.3M
Biogen	06/83	$57.5M
Chiron	08/83	$17M
Immunomedics	11/83	$2.5M
Repligen	04/86	$17.5M

Commercializing Biotechnology

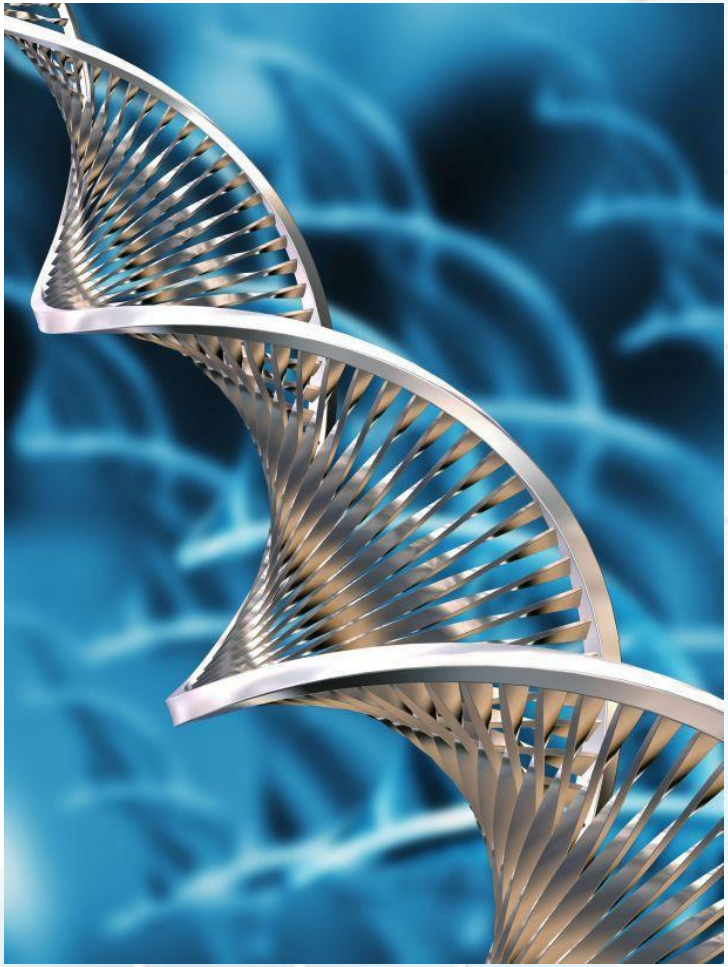
First generation of Biotech Companies

Company	Date of IPO	Amount Raised
OSI	04/86	$13.8M
Cytogen	06/86	$35.6M
Xoma	06/86	$32M
Genzyme	06/86	$28M
ImClone	06/86	$32M
Genetics Institute	05/86	$79M

Commercializing Biotechnology

Reference: from Alchemy to IPO (The business of Biotechnology) by Cynthia Robbins-Roth (2000)

Commercializing Biotechnology



Commercializing Biotechnology





- Roche acquired Genentech for $46.8 billion
- Current employees = 12,895
- 19 Breakthrough

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Biopharmaceutical Products

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Genentech
IN BUSINESS FOR LIFE

Roche



Commercializing Biotechnology

DR. HERBERT BOYER



- Revenue $20 billion
- Total Assets $69 billion
- Net Income $ 5 billion
- Current employees = 17,900
- 11 Breakthrough Biopharmaceutical

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Products

AMGEN



THE UNIVERSITY OF CALIFORNIA
LET THERE BE LIGHT
UCLA

DR. WINSTON SALSER

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Biogen

- Revenue $9.7 billion
- Total Assets $15 billion
- Net Income $ 3 billion
- Current employees = 7,550

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- 10 Breakthrough Biopharmaceutical Products



Commercializing Biotechnology

Biogen



THE UNIVERSITY OF EDINBURGH

Sir Kenneth Murray



UNIVERSITAS TURICENSIS MDCCC XXXIII

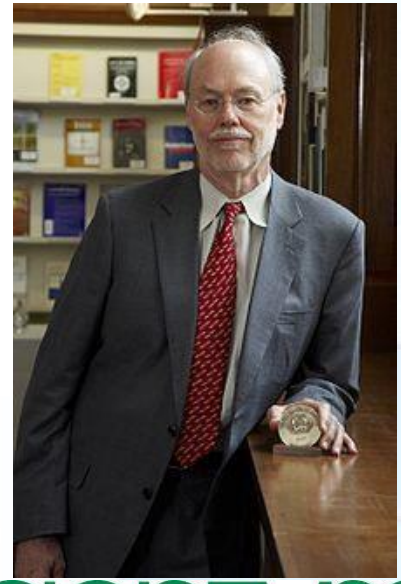
Commercializing Biotechnology

Charles Weissmann



Phillip Sharp

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genzyme

- Revenue $4.6 billion
- Net Income $ 421 million

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- Current employees = 12,000
- 16 innovative Biopharmaceutical Products



Gaucher's Disease

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Genetic disease in which fatty substances accumulate in cells and certain organs

Molecular Reason:

Hereditary deficiency of the enzyme glucocerebrosidase

Symptoms:

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Bruising, Fatigue, Anemia, low blood platelets, enlargement of the liver and spleen
Prior to this, they had to collect 22,000 placentas to treat Gaucher's patients

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genzyme



Tufts UNIVERSITY



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DR. HENRY BLAIR

genzyme



DR. HENRY BLAIR

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